NITIK JAIN

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Education (*expected)

Johns Hopkins University **Aug 2025 - May 2027***

MSE Robotics (Perception & Cognitive Systems)

- **Student Researcher, Johns Hopkins Medicine:** Deep Learning & AI for MRI enhancement and segmentation
- **Coursework:** Computer Vision, Deep Learning, Algorithms for Sensor-Based Robotics, Generative Models for Vision

Indian Institute of Technology (IIT), Kanpur **Jul 2017 - May 2022**

Integrated B.Tech + M.Tech, Mechanical & Aerospace Engineering; GPA: 9.4/10, *Minor in Machine Learning*

- **Master's Thesis:** Object Detection Underneath Sparse Occlusion of Forests *Advisor: Dr. Mangal Kothari*
- **Academic Excellence Award 2021** — best academic performance amongst all students in the dual-degree program

Work Experience

Lead AI Engineer — *Swiffylabs.ai, Bangalore, India* **Jun 2024 - Jun 2025**

WestBridge Capital-backed B2B fintech redefining core banking through applied AI; led AI development from 0→1

- Designed and deployed an AI-driven core banking stack, replacing legacy systems with **modular microservices**
- Integrated **LLM agents** and fraud-detection models into transaction, onboarding, and customer-experience workflows
- Partnered with NBFC clients to launch a **regulatory-ready, headless product suite**, leading the first commercial rollout

Founding AI Engineer — *Osfin.ai, Bangalore, India* **Apr 2023 - Jun 2024**

Sequoia-backed fintech building AI-first document intelligence; led product vision for an LLM-powered accounting suite

- Led cross-functional teams to define and ship strategic product roadmaps for **LLM-powered** document analysis
- Built **OCR**-based invoice extraction in Bahasa Indonesia, improving extraction accuracy by **40%**
- Developed an **LLM**-based email parser to extract structured data from unstructured mail across **30+** categories
- Engineered **ETL** pipelines into a **Vector DB** with clustering, reducing retrieval search space by **80%**

AI Scientist, R&D Division — *Qure.ai, Bangalore, India* **Sept 2022 – Mar 2023**

Deep Learning researcher for qCT-Chest — Qure.ai's flagship AI suite for early lung-cancer detection from CT scans

- Implemented a **3D ViT** in MONAI for nodule classification, reaching **88% AUC / 96% recall** — a **9%** gain over SoTA
- Augmented nodule **segmentation** with CBAM + SE layers in a U-Net backbone, improving the **DICE** coefficient by **14%**
- Built an end-to-end YAML-driven training framework on **PyTorch Lightning + Hydra** and scalable **Databricks** pipelines

Research Experience

AI Research Consultant — *MooVita Pte Ltd, Singapore* **Sept 2022 – Feb 2023**

Perception researcher in Computer Vision, focusing on explainability of visual perception for autonomous driving

- Leveraged **2D** features to build a **3D sparse point cloud** localized on monocular odometry derived from lane detection
- Engineered a novel fallback system using **classical optical flow** for rapid vehicle-turn estimation at **240 FPS** on CPU
- Devised an **object-detection fallback pipeline** using **RAFT's** dense optical flow, exploiting motion-based velocity cues

Sr. Student Researcher, IGCL — *IIT Kanpur* **Jun 2021 – Aug 2022**

Master's Thesis: Object Detection Underneath Sparse Occlusion of Forests   Advisor: Dr. Mangal Kothari

- Built a **sensor-fusion model** combining RGB and thermal modalities into integral images that reveal concealed targets
- Applied **Airborne Optical Sectioning (AOS)** in a synthetic-aperture framework, yielding ˜**70%** occlusion removal
- Fine-tuned **YOLOv5** via transfer learning on fused imagery, raising detection accuracy by **32%** over the base model

Undergraduate Researcher, IGCL — *IIT Kanpur* **Sept 2020 – May 2021**

Onboard 3D Detection and Tracking   Advisor: Dr. Mangal Kothari

- Engineered a real-time YOLOv3 detection-and-tracking pipeline optimized for speed and low resource usage on edge GPU
- Achieved **77% @ 0.5 IoU** on a **20 FPS** live feed, with tracking localization error within ˜**20%** up to 10 m

Visiting Research Intern, Robotics Research Center — *IIIT Hyderabad* **May 2019 – Jul 2019**

Motion Planning for Outdoor Autonomous UAVs   Advisor: Prof. K. Madhava Krishna

- Implemented **VINS-Mono** on ROS, fusing hardware-synced monocular camera, IMU, and **GPS-RTK** for robust state estimation
- Built a **voxel-based 3D mapper** from RealSense data, enabling real-time path planning via `mav_voxblox_planning`

SELECTED PROJECTS

Autonomous Ground Vehicle (AGV): Daksh II — Lead AI Researcher, Team VISiON, IIT Kanpur **Aug 2018 – May 2021**
*Built an outdoor autonomous ground vehicle; led the team at **IGVC 2019**, winning 4 international awards*

- Developed an end-to-end **autonomy stack in ROS**: Fast-SCNN segmentation, LiDAR–camera fusion mapping, and RRT planning

COVID Information Retrieval — Mentor: Prof. Arnab Bhattacharya, IIT Kanpur **Jan 2022 – Apr 2022** *Benchmarked IR frameworks for ranking CORD-19 research documents; published at Europe PMC*

- Compared **BM25** against **Contriever** and **BERT** embeddings on TREC-COVID, reaching a peak **MAP of 0.72**

SemEval-2021: Detecting & Rating Humour and Offence — Mentor: Prof. Piyush Rai **Sept 2020 – Dec 2020** *Sentiment models classifying short text as humorous, non-humorous, or offensive*

- Optimized **BERT-large** to **93%** accuracy and built a BERT+MLP humour-rating head with best **RMSE 0.446**

Parallelization of SLAM & Planning Algorithms — Mentor: Dr. Mangal Kothari **Aug 2019 – Nov 2019** *GPU-accelerated Particle-Filter SLAM on 2D LiDAR data*

- Parallelized **~87%** of pipeline steps on **CUDA**, achieving linear (vs. exponential CPU) scaling with particle count

ACADEMIC PUBLICATIONS & PATENTS

- **N. Jain**, P. Yadav, R. Das, I. Saha, *"Computer-Implemented Stigma Separation System"* *Indian Patent No. 564565 (2025)*
- **N. Jain**, *"Object Detection Underneath Sparse Occlusion of Forests"* *M.Tech Thesis, IIT Kanpur (2023)*
- **N. Jain**, M. Shukla, S. Gupta, *"IR Models and the COVID-19 Pandemic: A Comparative Study"* *Europe PMC, 2023*
- **N. Jain**, A. Shah, H. B. Reddy, M. Kothari, *"Design of a Low-Cost Autonomous Ground Vehicle"* *IEEE ICARSC 2022*
- **N. Jain**, A. Sharma, M. Kothari, *"Lightweight Multi-Drone Detection and 3D-Localization via YOLO"* *ArXiv 2022*

ACADEMIC SERVICE & TALKS

- **Teaching Assistant:** AE640A — Autonomous Systems (graduate course), IIT Kanpur *(Spring 2022)*
- **Invited Talk:** Basics of ML, Computer Vision & deploying CNNs in PyTorch — QIP Short-Term Course *(2021)*
- **Paper Reviewer:** Seventh Indian Control Conference (ICC-7) *(2021)*

AWARDS & ACHIEVEMENTS

- **J. N. Tata Scholar 2025** & **K. C. Mahindra Scholar 2025 —** among ~100 scholars worldwide each, for postgraduate study abroad *(2025)*
- **SIIC Student Innovation Award** for the Saffron Stigma Separator; **Amazon AWS ML Scholarship** *(2022 / 2021)*
- **Lescoe Cup — 2nd Worldwide** & **3rd AutoNav**, 27th Intelligent Ground Vehicle Competition, Michigan, USA *(2019)*
- **Silver** (TCTD, 7th Inter-IIT) & **Bronze** (ATC, 6th Inter-IIT) medals; **All-India Rank 1116** in JEE Advanced (1.5M candidates) *(2017–19)*

LEADERSHIP

- **Team Leader, Team VISiON (ex-IGVC), IIT Kanpur —** led a 3-tier team of 21 students in AI & Robotics R&D; raised **$35,000** in funding for a low-cost autonomous EV and secured an NVIDIA Academic Hardware Grant ($3,000 in GPUs) *(2019–2021)*

TECHNICAL SKILLS

Languages: Python *(Proficient)*, C/C++, MATLAB, SQL, Bash, R

ML / DL: PyTorch, PyTorch Lightning + Hydra, MONAI, Hugging Face, LangChain + LangSmith, Scikit-learn, OpenCV

Robotics & Tools: ROS + Gazebo, Docker, Databricks, Grafana, CloudWatch, Git, Autodesk Inventor, LaTeX